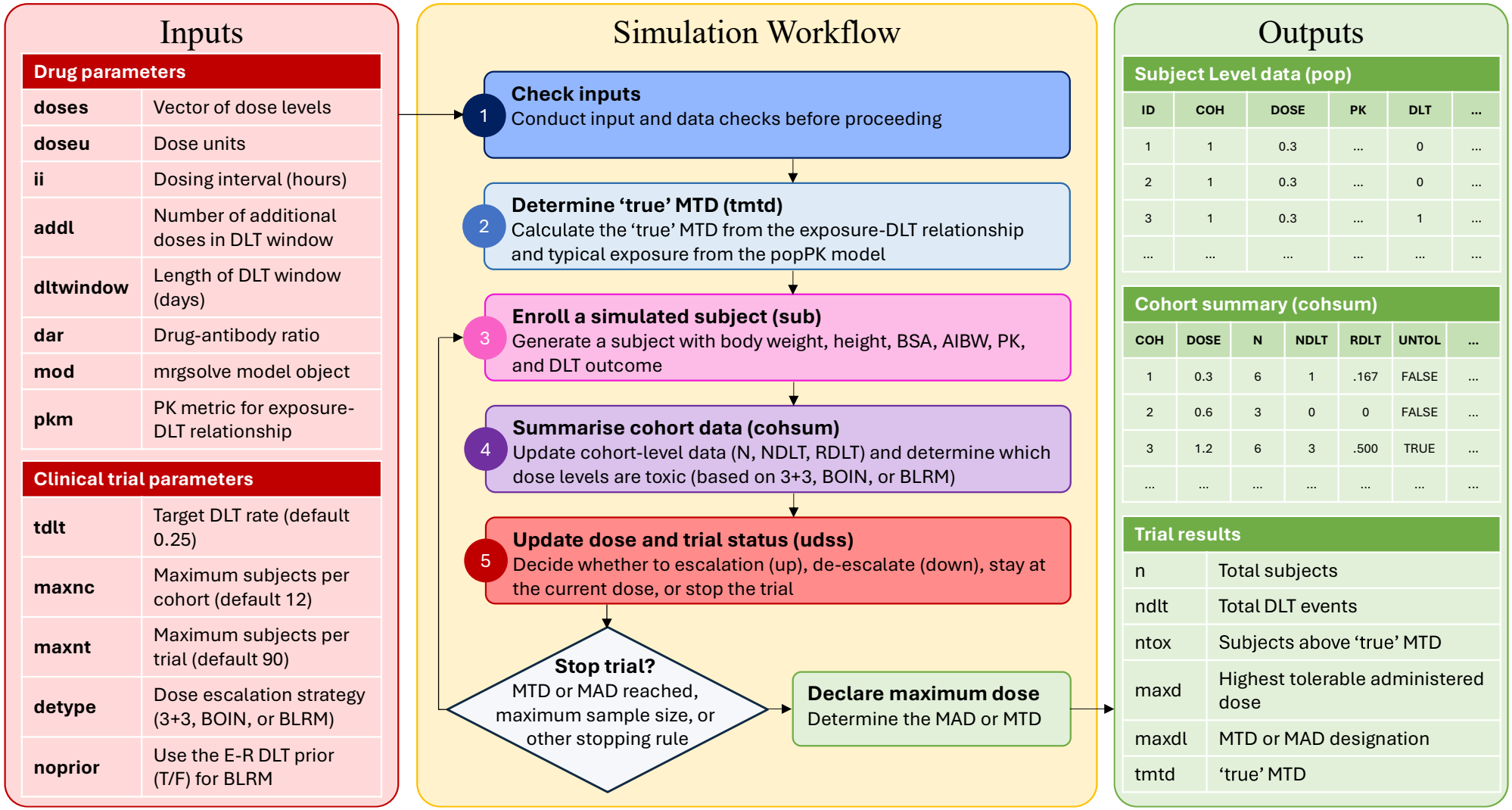

Inputs
Drug parameters
doses
Vector of dose levels
doseu
Dose units
ii
Dosing interval (hours)
addl
Number of additional doses in DLT window
dltwindow
Length of DLT window (days)
dar
Drug-antibody ratio
mod
mrgsolve model object
pkm
PK metric for exposure-DLT relationship
Clinical trial parameters
tdlt
Target DLT rate (default 0.25)
maxnc
Maximum subjects per cohort (default 12)
maxnt
Maximum subjects per trial (default 90)
detype
Dose escalation strategy (3+3, BOIN, or BLRM)
noprior
Use the E-R DLT prior (T/F) for BLRM
Simulation Workflow
1
Check inputs
Conduct input and data checks before proceeding
2
Determine 'true' MTD (tmtd)
Calculate the 'true' MTD from the exposure-DLT relationship and typical exposure from the popPK model
3
Enroll a simulated subject (sub)
Generate a subject with body weight, height, BSA, AIBW, PK, and DLT outcome
4
Summarise cohort data (cohsum)
Update cohort-level data (N, NDLT, RDLT) and determine which dose levels are toxic (based on 3+3, BOIN, or BLRM)
5
Update dose and trial status (udss)
Decide whether to escalation (up), de-escalate (down), stay at the current dose, or stop the trial
Stop trial?
MTD or MAD reached, maximum sample size, or other stopping rule
Declare maximum dose
Determine the MAD or MTD
Outputs
Subject Level data (pop)
ID COH DOSE PK DLT ...
1 1 0.3 ... 0 ...
2 1 0.3 ... 0 ...
3 1 0.3 ... 1 ...
... ...
Cohort summary (cohsum)
COH DOSE N NDLT RDLT UNTOL ...
1 0.3 6 1 .167 FALSE ...
2 0.6 3 0 0 FALSE ...
3 1.2 6 3 .500 TRUE ...
... ...
Trial results
n Total subjects
ndlt Total DLT events
ntox Subjects above 'true' MTD
maxd Highest tolerable administered dose
maxdl MTD or MAD designation
tmtd 'true' MTD